APPENDIX A
IMPLEMENTATION DETAILS

A. Network architectures

Both policies are trained with PPO [1] using an actor–critic architecture in which the policy and value functions share a common observation encoder. For each task, the same network architecture is used for both robotic platforms.

Navigation policy. The single-channel 200×150 ultrasound image is processed by a four-layer convolutional encoder. The layers have 32, 64, 64, and 64 output channels; kernel sizes 8×8, 4×4, 4×4, and 3×3; and strides 2, 2, 2, and 1, respectively. Each convolutional layer is followed by a ReLU activation. The flattened image features are then processed by an MLP with hidden sizes 512, 256, and 128 and ELU activations. Separate linear heads output the mean of the 3-D action distribution and the scalar value estimate.

Drilling policy. The $7 \times 200 \times 150$ ultrasound stack is processed by treating the seven slices as input channels. The convolutional encoder comprises four layers with 32, 64, 64, and 64 output channels; kernel sizes 8×8, 4×4, 3×3, and 4×4; and strides 4, 2, 1, and 2, respectively. Each convolutional layer is followed by a ReLU activation. The relative drill–probe position and orientation are processed by two separate MLPs, each with hidden sizes 128, 128, and 64 and ELU activations. The resulting image and pose features are concatenated and passed through a shared MLP with hidden sizes 512, 256, and 128 and ELU activations. Separate linear heads output the mean of the action distribution and the scalar value estimate. For both policies, the action log-standard deviation is represented by a learned, state-independent parameter.

B. Training Details

All policies were trained on a workstation equipped with an NVIDIA RTX 4090 GPU. Table I reports the approximate wall-clock time required for each training stage, patient model, and embodiment. Across the five patient models and two embodiments, the ten complete pipelines required approximately 222 h. Training parameters values are reported in Tables II– V.

TABLE I
WALL-CLOCK TIME PER PATIENT MODEL AND EMBODIMENT (H:MIN).

Stage	Unitree G1	KUKA LBR Med14
Navigation training	04:30	01:50
Drilling training	10:10	06:05
Drilling pre-training	05:00	03:10
CEM probe–target optimization	01:15	01:00
Drilling fine-tuning	03:50	01:15
Total per patient model	29:15	15:10

APPENDIX B
EXTENDED ANALYSIS

A. Per-patient results

Table VI reports drilling and navigation performance separately for each patient model, embodiment, and pipeline.

TABLE II
PPO HYPERPARAMETERS FOR THE FOUR TRAINING STAGES.

Parameter	Nav.	Drill	Pre-tr.	Fine-t.
Parallel environments		64		
Rollout length	32	16	16	32
Trainer timesteps	2×10^5	2×10^5	2×10^5	8×10^4
Optimizer		Adam		
Learning rate	10^{-4}	10^{-4}	10^{-4}	3×10^{-5}
Epochs per update	5	3	3	2
Minibatches per update	32	32	32	8
KL scheduler threshold	0.008	0.008	0.008	0.004
Maximum gradient norm	1.0	1.0	1.0	0.5
Discount factor γ		0.99		
GAE coefficient λ		0.95		
PPO clip range		0.20		
Value-function clip range	0.20	0.10	0.10	0.05
Entropy coefficient		0		
Value-loss coefficient		1.0		

TABLE III
REWARD PARAMETERS. WEIGHT VECTORS ARE ORDERED AS (w_l, w_r, w_a).

Task/stage	Parameter	Unitree G1	KUKA LBR Med14
Navigation	w_{uv}	0.06	0.03
	w_ϕ	1	1
	λ_{stab}	0.25	-
	r_{succ}	0.5	-
	N_{hold}	1 s	-
	Radial tolerance	5 mm	-
	Angular tolerance	5°	-
Drilling	$\mathbf{w}^{\text{FS}}$	$(30, 60, 4)$	$(30, 30, 5)$
	$\mathbf{w}^{\text{SC}}$	$(600, 300, 30)$	$(450, 300, 15)$
	$\mathbf{w}^{\text{U}}$	$(30, 60, 4)$	$(30, 30, 5)$
	λ_u	20	0
	SC radial tolerance	15.5 mm	9.5 mm
	SC angular tolerance	5°	5°
Drilling fine-tuning	$\mathbf{w}^{\text{FS}}$	$(30, 60, 4)$	$(30, 30, 5)$
	$\mathbf{w}^{\text{SC}}$	$(150, 600, 30)$	$(225, 300, 15)$
	$\mathbf{w}^{\text{U}}$	$(30, 60, 4)$	$(30, 30, 5)$
	λ_u	20	0
	SC radial tolerance	15.5 mm	9.5 mm
	SC angular tolerance	5°	5°

B. Ablations

We evaluate three conditions to analyze the contribution of probe-target optimization and the behavior of the drilling policy under active navigation. Condition **A** corresponds to the complete *dual-policy optimization* pipeline described in the paper. Condition **B** follows the same procedure but performs the fine-tuning stage using the navigation policy trained for the nominal probe target instead of the optimized one. Condition **C** evaluates the randomized pre-trained drilling policy before fine-tuning, while the probe is held fixed at the optimized target pose.

Comparing conditions **A** and **B** isolates the contribution of the optimized probe target within the complete pipeline, since both conditions use active navigation and the same drilling-policy initialization and fine-tuning procedure. Condition **C** evaluates the capability of the pre-trained drilling policy under

TABLE IV
ACTION SCALING VECTORS. TRANSLATIONAL COMPONENTS PRECEDE ROTATIONAL COMPONENTS: TWO AND ONE FOR NAVIGATION, AND THREE AND TWO FOR DRILLING, RESPECTIVELY.

Task	Embodiment	Action scale vector
Navigation	Unitree G1	(0.15, 0.15, 0.015)
	KUKA LBR Med14	(0.50, 0.50, 0.050)
Drilling	Unitree G1	(0.01, 0.01, 0.01, 0.10, 0.10)
	KUKA LBR Med14	(0.10, 0.10, 0.10, 1.00, 1.00)

TABLE V
CEM PARAMETERS FOR PROBE–TARGET OPTIMIZATION. COMPONENTS OF $\boldsymbol{\mu}_0$ AND $\boldsymbol{\sigma}_0$ ARE ORDERED AS (u, v, ϕ), WITH UNITS $(\mathrm{mm}, \mathrm{mm}, \mathrm{rad})$.

Parameter	Value
Population size N	28
Elite fraction ρ	0.20
CEM iterations K	12
Rollouts per candidate N_{roll}	64
Update coefficient α	0.75
Initial mean $\boldsymbol{\mu}_0$	$(-45, 0, 1.57)$
Initial std. $\boldsymbol{\sigma}_0$	$(24, 24, 0.16)$
Search range u	$[-82.5, -15]\,\mathrm{mm}$
Search range v	$[-37.5, 37.5]\,\mathrm{mm}$
Search range ϕ	$[76, 101]^\circ$
Safety weight λ_{safe}	0.5
SC radial tolerance	$7.5\,\mathrm{mm}$
SC angular tolerance	10°

fixed imaging at the optimized target, allowing comparison with condition **A** to assess the benefits of the complete workflow, including fine-tuning under active navigation, over this pre-trained starting point.

Fine-tuning at the nominal target (**B**) yields worse performance than the complete pipeline (**A**), with more pronounced degradation on the G1. This indicates that optimized probe-target selection is especially important for the G1 within the complete workflow.

The pre-trained drilling policy (**C**) also performs substantially worse than the complete pipeline, despite the probe being fixed at the optimized target. As discussed in the paper, the broad randomization of probe poses during pre-training makes it difficult to achieve accurate drilling at this stage. Its primary role is to support the selection of a promising probe target, providing an initialization for subsequent fine-tuning rather than a fully refined drilling policy.

C. Probe-target optimization and manipulability

We investigate whether CEM selects probe targets primarily because they are kinematically favorable, or whether the resulting ultrasound view also contributes to the improvement. We focus this analysis on the Unitree G1, whose shared torso joints couple the simultaneous control of the probe and drill, introducing kinematic constraints that are absent in the dual-KUKA setup. Previous work has used manipulability-based reachability maps to optimize robot base placement and probe-holder geometry in robotic ultrasound [2], motivating an examination of the kinematic factors underlying probe-target selection.

For each patient model, we initialize the probe at a randomly sampled configuration within the probe-target search region and position the drill tip along the planned insertion trajectory, approximating kinematic conditions encountered during the intervention. We then apply random probe-motion commands while maintaining the drill target fixed. To simplify the analysis, we hold the commanded in-plane probe orientation at 1.57 rad. At each simulation step, we compute the Yoshikawa manipulability index from the Jacobian of the probe-supporting wrist frame, including all active upstream joints of the torso and probe arm, and associate it with the measured probe position. Samples outside the search region or exceeding the drill-pose error tolerances are discarded. Measurements are averaged within spatial cells for each environment and then across environments. The resulting maps are interpolated and smoothed for visualization, without extrapolation beyond the sampled support.

Fig. 1 compares these maps with the CEM objective over five patient models. Qualitatively, regions with high objective values do not consistently coincide with regions of high manipulability, and the CEM-selected targets do not systematically maximize manipulability. This suggests that manipulability alone does not explain the spatial variation of the CEM objective and is consistent with a contribution from the ultrasound view. This qualitative comparison does not, however, isolate the contributions of imaging and kinematics.

D. Probe error during the drilling phase

To better understand the effects of dual-policy optimization, we analyze navigation performance during drilling. We restrict this analysis to the G1 humanoid, where navigation and drilling are kinematically coupled through the shared waist joints. We measure the planar probe position error relative to the target scanning configuration, aggregating rollouts across the five patient models for each pipeline. Figure 2 shows the complete positioning and drilling sequence, while Fig. 3 focuses on the drilling episode.

Navigation performance is not substantially degraded by simultaneous drilling, as the mean error remains bounded and generally decreases or stabilizes during the episode. The small transient increase shortly before drilling begins (Fig. 2) is associated with moving the drill to its randomized initial pose, which briefly perturbs probe tracking.

The relative performance of the two pipelines is consistent with that observed during isolated navigation, with dual-policy optimization maintaining lower probe errors. This suggests that its improvement in surgical performance may partly stem from more accurate navigation, although this comparison alone does not establish a causal relationship.

E. Distribution of achieved insertion depth

Fig. 4 shows the insertion depth distributions separately for the five patient models, both embodiments, and both settings.

TABLE VI

PER-PATIENT DRILLING AND NAVIGATION RESULTS. VALUES ARE MEANS OVER 100 EVALUATION ROLLOUTS FOR EACH PATIENT MODEL AND EMBODIMENT.

Setting	Embodiment	Patient	Radial err. [mm]	Angular err. [deg]	Insertion depth [%]	SR [%]	Nav. pos. err. [mm]	Nav. ang. err. [deg]
Baseline	KUKA LBR Med14	p10	1.77	0.74	71.3	98.44	2.26	1.47
		p15	1.67	0.88	74.6	100.0	1.36	1.12
		p28	0.83	0.47	80.1	98.54	0.46	0.43
		p30	2.01	0.86	64.8	100.0	1.48	1.00
		p34	2.24	1.00	73.5	98.57	1.86	1.04
	Unitree G1	p10	3.67	2.44	59.5	98.8	3.09	2.22
		p15	4.37	4.15	74.5	95.9	3.01	1.68
		p28	2.29	3.66	67.0	97.5	3.44	1.46
		p30	3.89	4.59	42.1	97.8	3.11	2.05
		p34	3.90	4.70	62.3	99.7	4.09	3.30
Dual-policy optimization	KUKA LBR Med14	p10	0.70	1.02	77.9	100.0	0.98	0.48
		p15	1.40	0.28	70.7	99.8	0.88	0.32
		p28	0.88	0.55	80.6	100.0	0.60	0.33
		p30	1.17	0.97	75.3	99.9	0.95	0.30
		p34	0.73	0.93	76.1	99.8	1.24	0.59
	Unitree G1	p10	1.76	4.02	64.2	100.0	2.13	3.09
		p15	3.54	1.48	65.6	100.0	1.79	2.42
		p28	1.41	1.74	68.2	98.46	1.67	1.11
		p30	2.88	3.97	67.6	97.99	1.87	0.85
		p34	2.02	1.80	71.3	100.0	3.13	1.86

TABLE VII

ABLATION OF THE PROBE-TARGET OPTIMIZATION STAGES. EACH CONDITION IS EVALUATED OVER 100 ROLLOUTS PER PATIENT MODEL. MEANS AND STANDARD DEVIATIONS ARE COMPUTED OVER ALL EVALUATION ROLLOUTS POOLED ACROSS THE FIVE PATIENT MODELS.

Setting	**Embodiment**	**Radial err. [mm]** $\downarrow$	**Angular err. [deg]** $\downarrow$	**Insertion depth [%]** $\uparrow$	**SR [%]** $\uparrow$
A	KUKA LBR Med14	0.97 ± 0.64	0.75 ± 0.64	76.04 ± 3.38	99.92
	Unitree G1	2.32 ± 1.19	2.58 ± 1.20	67.35 ± 3.36	99.28
B	KUKA LBR Med14	2.35 ± 1.91	0.83 ± 0.74	59.88 ± 7.91	97.42
	Unitree G1	5.59 ± 7.25	4.54 ± 7.38	38.12 ± 26.78	84.10
C	KUKA LBR Med14	4.47 ± 1.73	2.09 ± 2.04	71.10 ± 19.34	97.46
	Unitree G1	7.03 ± 8.13	6.70 ± 9.12	51.88 ± 18.65	70.85

Depth is expressed as a percentage of the maximum insertion depth. The distributions remain below full insertion and generally exhibit a single dominant mode, although several settings show asymmetric tails or smaller secondary peaks. Comparing the baseline workflow with dual-policy optimization, the dominant mode shifts towards greater insertion depth for p10, p30, and p34 in both embodiments, whereas the distributions largely overlap for p28. For p15, the optimized distributions instead shift towards shallower insertion. Across platforms, the dual-KUKA LBR Med14 distributions are typically more concentrated around their dominant mode, whereas Unitree G1 exhibits broader distributions and more pronounced variation across patient models. Some patients exhibit secondary peaks or asymmetric tails, reflecting heterogeneity across rollouts. The different stopping depths emerge from the learned policies rather than an explicitly prescribed intermediate target. Partial insertion may reflect a trade-off between further advancement and maintaining trajectory accuracy and safety; however, the present analysis does not establish the mechanism responsible for this behavior.

REFERENCES

[1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[2] J. Osburg, N. T. Nguyen, and F. Ernst, "Increasing reachability in robotic ultrasound through base placement and tool design," *The International Journal of Medical Robotics and Computer Assisted Surgery*, vol. 20, p. e70037, 2024.

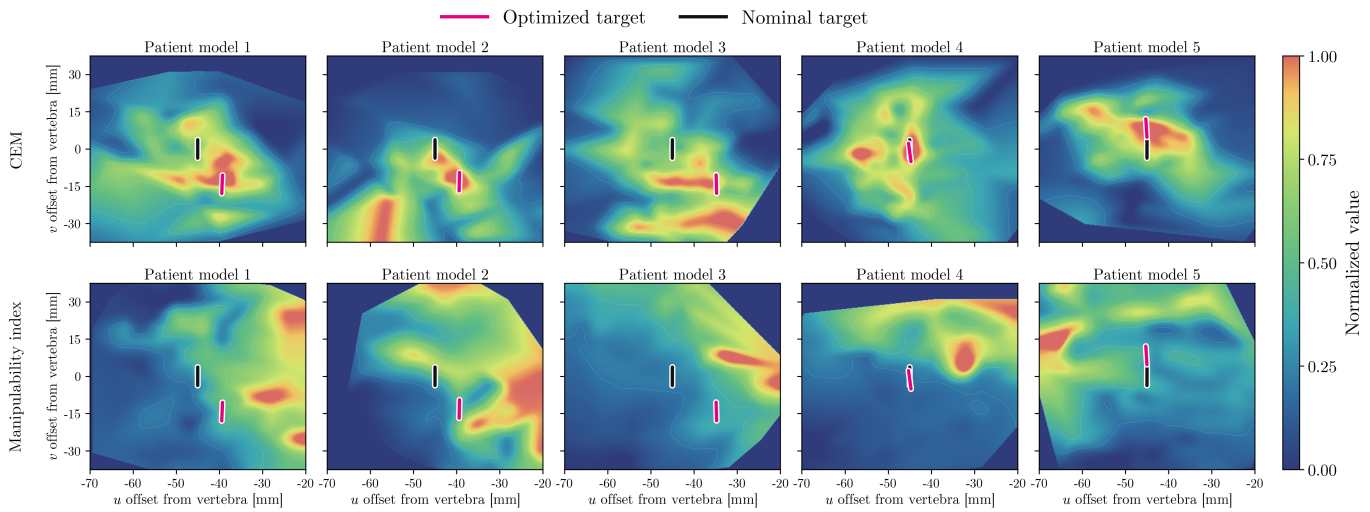


Fig. 1. CEM objective (top) and probe manipulability (bottom) for the Unitree G1 across five patient models. Columns correspond to patients p10, p15, p28, p30, and p34. Each panel is independently normalized for visualization. Magenta and black markers indicate the CEM-selected and nominal probe targets, respectively; the same targets are shown in both rows. Unsupported regions are displayed in dark blue but are excluded from the analysis.

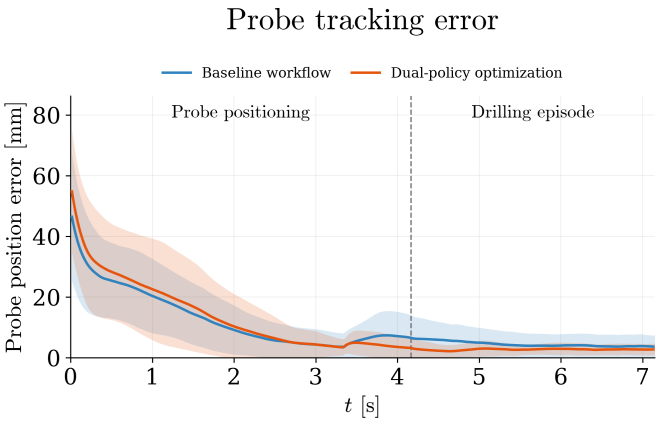


Fig. 2. Probe position error during positioning and drilling on G1. Curves show the mean across rollouts from five patient models for each pipeline; shaded regions indicate ± 1 standard deviation. The dashed line marks the start of drilling.

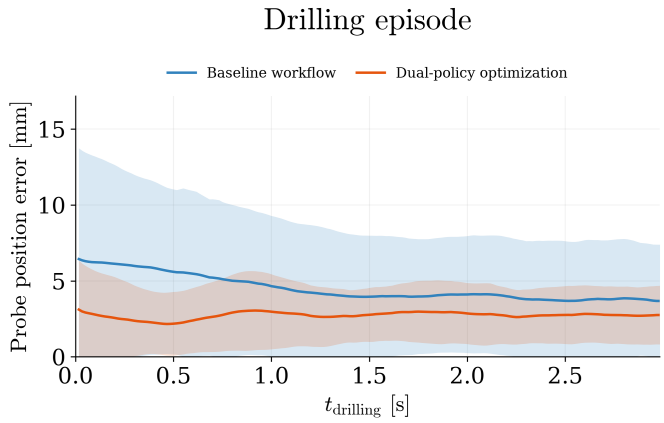


Fig. 3. Probe position error during drilling on G1, with time measured from drilling onset. Curves and shaded regions show the mean and ± 1 standard deviation across rollouts from five patient models for each pipeline.

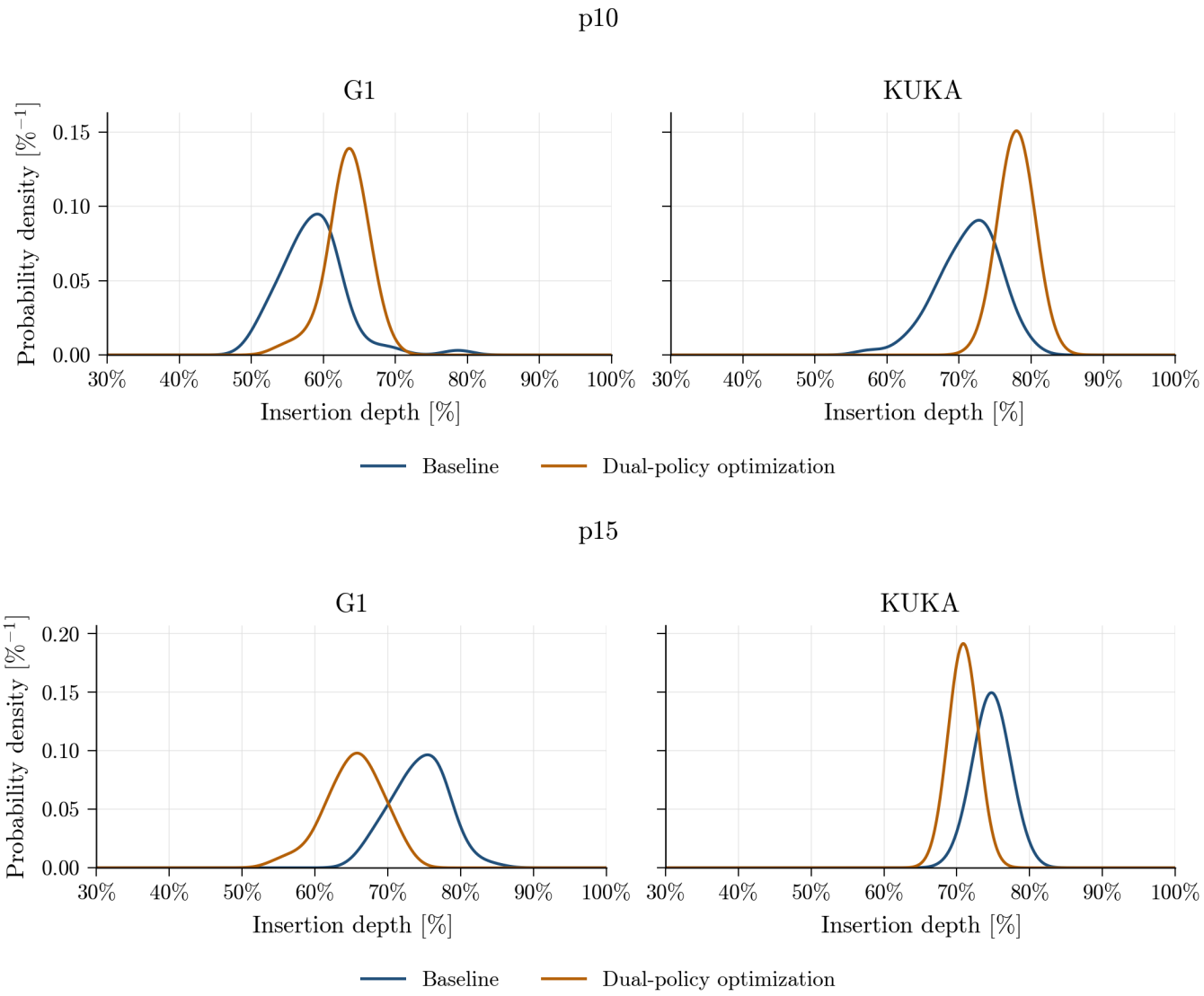


Fig. 4. Gaussian kernel density estimates of insertion depth as a percentage of the prescribed maximum for patients p10 and p15. Left and right panels show Unitree G1 and dual-KUKA LBR Med14, respectively. Blue denotes the baseline and orange denotes dual-policy optimization. Distributions are shown separately for each patient, without pooling.

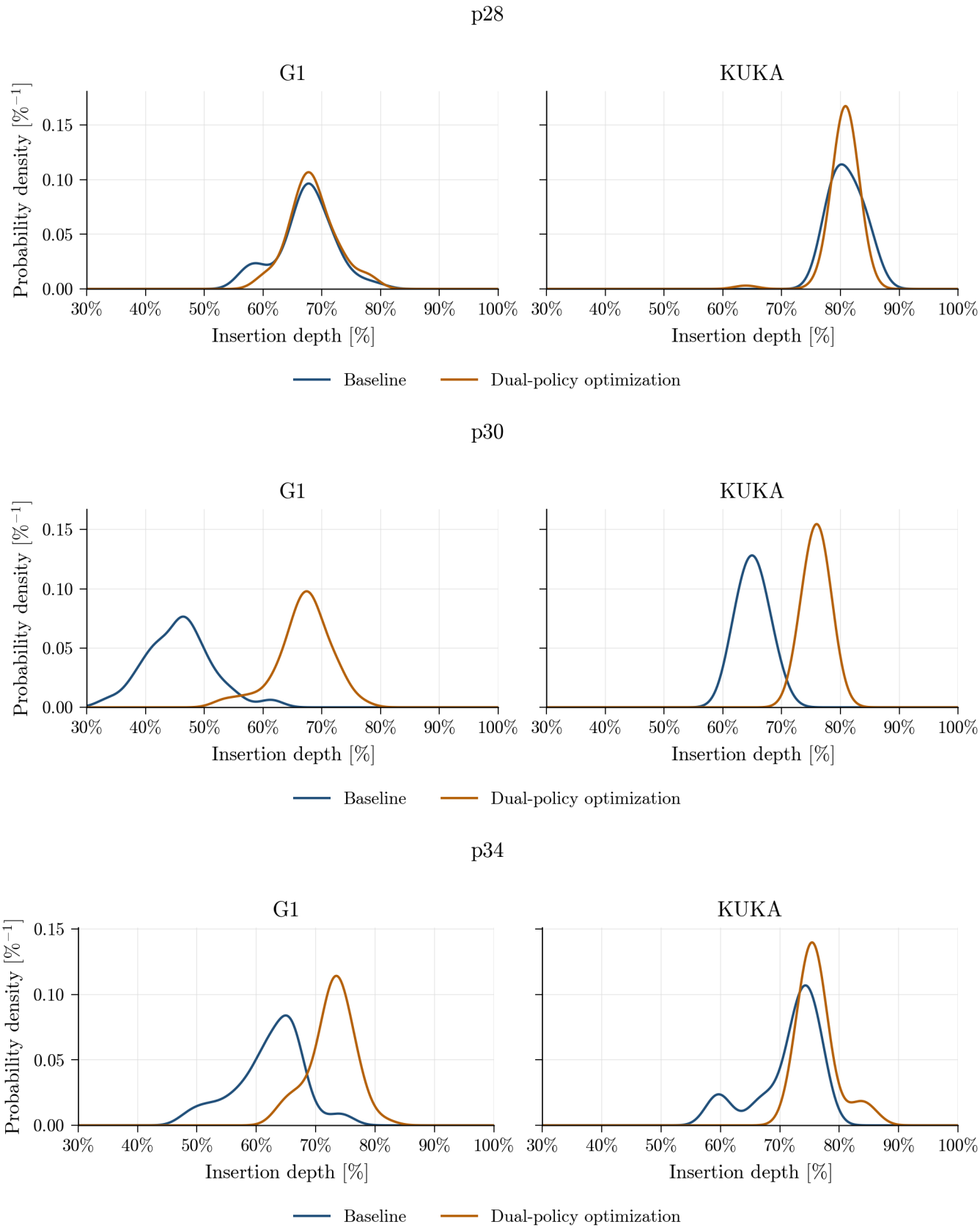


Fig. 4. Insertion depth distributions, continued: patients p28, p30, and p34. Panel arrangement and colors are as in the preceding figure.